

Program: Diploma in Cloud Computing and Big Data	
Course Code: 5278	Course Title: Big Data Lab
Semester: 5	Credits: 1.5
Course Category: Programme core course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Optimize decisions and improve efficacy in business administration with Big data analytics
- Practice java concepts required for developing map reduce programs.
- Impart the architectural concepts of Hadoop and introducing map reduce paradigm
- Familiarize programming tools PIG and HIVE in the Hadoop ecosystem.
- Implement best practices for Hadoop development in real world problems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic programming concepts in cloud	3276	Cloud computing Lab	III
Operating systems concepts	3278	Linux System Administration Lab	III
Map Reduce programming in Hadoop Environment & basic concepts of Hive and Pig	4271	Big Data Concepts	IV
Sound knowledge in Python Programming	1008	Introduction to IT systems Lab Programming in Python	1
basic concepts of Database Management Systems	3272	Database System Concepts	III

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the installation of VMWare & Hadoop operating modes.	8	Applying
CO2	Practice the basic commands of LINUX Operating System & file management in Hadoop.	6	Applying
CO3	Illustrate Map Reduce Paradigm & its programming applications in solving real time problems.	14	Applying
CO4	Perform the installation and applications of programming tools like PIG & HIVE in the real world Hadoop Ecosystem.	14	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	2	2					
CO3	3	3					
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the installation of VMWare & Hadoop operating modes.		
M1.01	Demonstrate the installation of VMWare to setup the Hadoop environment and its ecosystems	2	Applying

M1.02	Demonstrate setting up and Installing Hadoop in its three operating modes. i. Standalone. ii. Pseudo distributed. iii Fully distributed.	4	Applying
M1.03	Use web-based tools to monitor your Hadoop setup.	2	Applying
Contents: Installation of VMWare to setup the Hadoop environment and its ecosystems, setting up and Installing Hadoop in its three operating modes- Standalone, Pseudo distributed, fully distributed, use web-based tools like spark,impala,mahout, Chukwa,Apache HBase,Ambari etc to monitor your Hadoop setup.			
CO2	Practice the basic commands of LINUX Operating System & file management in Hadoop.		
M2.01	Practice the basic commands of LINUX Operating System – File/Directory creation, deletion, update operations- use commands like pwd,mkdir,rmdir,ls,cd,touch,cat,rm,cp,mv etc	3	Applying
M2.02	Implement the following file management tasks in Hadoop: i. Adding files and directories ii. Retrieving files iii. Deleting files Hint: A typical Hadoop workflow creates data files (such as log files) elsewhere and copies them into HDFS using one of the above command line utilities.	3	Applying
	Lab Exam – I	1	

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the installation of VMWare & Hadoop operating modes.		
M1.01	Demonstrate the installation of VMWare to setup the Hadoop environment and its ecosystems	2	Applying
M1.02	Demonstrate setting up and Installing Hadoop in its three operating modes. i. Standalone. ii. Pseudo distributed. iii Fully distributed.	4	Applying
M1.03	Use web-based tools to monitor your Hadoop setup.	2	Applying
Contents: Use commands like pwd,mkdir,rmdir,ls,cd,touch,cat,rm,cp,mv,etc for practicing basic programs with linux operating systems.File Management tasks in Hadoop- Create directory in HDFS at given path(s), List the contents of directory, Upload and download file in HDFS, View contents of file, copy file from source to destination, copy file from/To Local file system to HDFS, move file from source to destination, remove file or directory in HDFS, display last few lines of a file, Rename File/Directory, Display the aggregate length of a file			
CO3	Illustrate Map Reduce Paradigm & its programming applications in solving real time problems.		
M3.01	Practice basic word count Map Reduce program to understand Map Reduce Paradigm.	2	Applying
M3.02	Write a Map Reduce program that mines weather data. Hint: Weather sensors collecting data every hour at many locations across the globe gather a large volume of log data, which is a good candidate for analysis with Map Reduce, since it is semi structured and record-oriented	3	Applying
M3.03	Practice matrix multiplication with Hadoop Map Reduce.	2	Applying
M3.04	Apply K-means clustering using Map Reduce	3	Applying
M3.05	Make use of Open Ended Experiments** to solve real time problems.	4	Applying

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CO1	Demonstrate the installation of VMWare & Hadoop operating modes.		
M1.01	Demonstrate the installation of VMWare to setup the Hadoop environment and its ecosystems	2	Applying
M1.02	Demonstrate setting up and Installing Hadoop in its three operating modes. i. Standalone. ii. Pseudo distributed. iii Fully distributed.	4	Applying
M1.03	Use web-based tools to monitor your Hadoop setup.	2	Applying
Contents: Set up Map Reduce Paradigm to Run a basic word count, mines weather data, matrix multiplication & K-means clustering. ** - Sample Open Ended Experiments (Not evaluated for End Semester Examination but to be included in Continuous Internal Evaluation.) Students can-do open-ended real time projects/ experiments as a group of 2-3. There is no duplication in experiments between groups. Consider the following sample situations: • Twitter Data Sentiment Analysis • Wiki Page Ranking with Hadoop • Health Care Data Management using Apache Hadoop Ecosystem • Retail Data Analysis Using Big Data • Aadhar Based Analysis using Hadoop • Climatic Data Analysis Using Hadoop • Flight History Analysis • Pseudo-Distributed Hadoop Cluster in Script • Sensex log data processing using Big Data tools Students should gather the required information, develop algorithms, map them to the Hadoop ecosystems to implement.			
CO4	Perform the installation and applications of programming tools like PIG & HIVE in the real world Hadoop Ecosystem.		
M4.01	Illustrate the installation of PIG.	2	Applying

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CO1	Demonstrate the installation of VMWare & Hadoop operating modes.		
M1.01	Demonstrate the installation of VMWare to setup the Hadoop environment and its ecosystems	2	Applying
M1.02	Demonstrate setting up and Installing Hadoop in its three operating modes. i. Standalone. ii. Pseudo distributed. iii Fully distributed.	4	Applying
M1.03	Use web-based tools to monitor your Hadoop setup.	2	Applying
M4.02	Write Pig Latin scripts sort, group, join, project, and filter your data. Demonstrate how to a. Run the Pig Latin Scripts to find Word Count b. Run the Pig Latin Scripts to find a max temp for each and every year.	3	Applying
M4.03	Illustrate the installation of HIVE.	2	Applying
M4.04	Use Hive to create, alter, and drop databases, tables, views, functions, and indexes.	2	Applying
M4.05	Open Ended Experiments**	5	Applying
	Lab Exam – II	2	

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the installation of VMWare & Hadoop operating modes.		
M1.01	Demonstrate the installation of VMWare to setup the Hadoop environment and its ecosystems	2	Applying
M1.02	Demonstrate setting up and Installing Hadoop in its three operating modes. i. Standalone. ii. Pseudo distributed. iii Fully distributed.	4	Applying
M1.03	Use web-based tools to monitor your Hadoop setup.	2	Applying
Contents: Set up the installation procedures of PIG & HIVE. Write Pig Latin scripts sort, group, join, project, and filter your data. Demonstrate how to (a) Run the Pig Latin Scripts to find Word Count (b) Run the Pig Latin Scripts to find a max temp for each and every year. Use Hive to create, alter, and drop databases, tables, views, functions, and indexes. ** - Sample Open Ended Experiments (Not evaluated for End Semester Examination but to be included in Continuous Internal Evaluation.) Students can-do open-ended real time projects/ experiments as a group of 2-3. There is no duplication in experiments between groups. Consider the following sample scenarios: • Corporate data integration. • A use case for scalability. • Facebook Data Analysis Using Hadoop and Hive • Link prediction for social media sites. • Document analysis application. • Specialized analytics. • Streaming analytics. • Web Based Data Management of Apache Hive Students should gather the required information, develop algorithms, map them to the Hadoop ecosystems to implement.			

Text / Reference

T/R	Book Title/Author
T1	Jay Liebowitz, —Big Data and Business Analytics Laboratory, CRC Press.
T2	Seema Acharya, Subhashini Chellappan, "Big Data Analytics", Wiley, 2015
R1	Tom White, — Hadoop: The Definitive Guide Third Edition, O'Reilly Media, 2012

Online Resources

Sl.No	Website Link
1	https://www.vmware.com/tryvmware/?p=workstation-w
2	https://www.tutorialspoint.com/apache_pig/index.htm
	https://www.tutorialspoint.com/hive/index.htm
3	http:// apache.mirrors.lucidnetworks.net/hadoop/
3	https://hadoop.apache.org/docs/r1.2.1/mapred_tutorial.html